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Concept Review
Stepper Motors

Why Use Steppers?

A stepper motor is a brushless, synchronous electric motor that moves in discrete steps. Unlike DC or BLDC motors, which rotate continuously when powered, a stepper motor divides a full rotation into a number of equal steps. A stepper motor is driven by applying timed electrical pulses that rotate it by fixed increments, making it ideal for precise position and speed control without requiring feedback sensors. Stepper motors are commonly used in applications like 3D printers, CNC machines, and robotics where accuracy and repeatability are more important than raw speed or torque.

Introduction

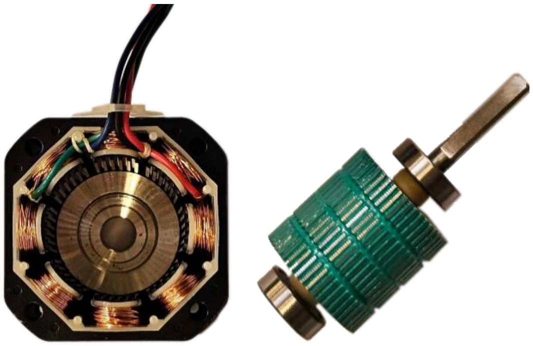
For many mechatronic applications, it is necessary to be able to control the position of an actuator with precision. In contrast to the continuous rotation of a brushed or brushless DC motor, a stepper motor rotates the armature in discrete steps. The fixed coil spacing in the motor produces rotation in consistent step sizes and maintain high holding torque. As the motor is only stepping between known distinct positions, it can be used for holding through only open-loop control without the need for sensor feedback. This is one feature that distinguishes the stepper motor from brushed and brushless DC motors which utilize closed-loop position control.

Basic Construction and Drive

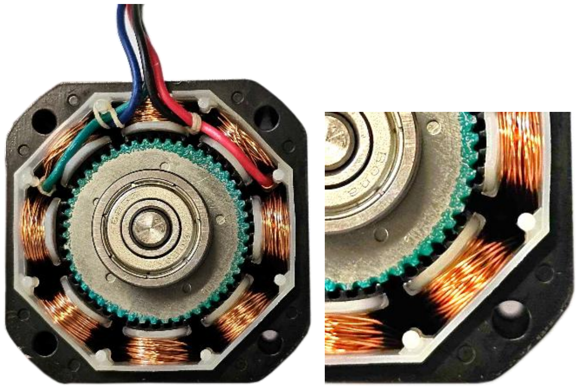
A stepper motor consists of two parts. The **rotor** is the rotating part which is typically a permanent magnet or a toothed iron core that reacts to the magnetic fields generated by the stator. The **stator** contains multiple windings arranged in phases, which are energized in a specific sequence to produce a rotating magnetic field. As the field rotates, the rotor aligns with it, stepping forward with each pulse of current.

The number of phases in a stepper motor refers to the number of separate coils it contains, which are energized together to produce a magnetic field. The number of steps per revolution is determined by the internal design of the motor—common stepper motors have a step angle of 1.8°, resulting in 200 steps per full rotation.

The stepper motor that is provided with the Mechatronic Actuators Trainer is a hybrid stepper motor that has a permanent magnet toothed rotor and a toothed stator. Figure 1 (a) shows the two different pieces, the stator and the rotor and Figure 1 (b) shows the inside of an assembled stepper motor and how some teeth from the rotor align with the stator while others do not.



(a). Stator with 8 coils and rotor



(b) Assembled stepper and teeth alignment

Figure 1: Interior of a stepper motor

The stepper motor that is provided with the Mechatronic Actuators Trainer has 8 poles that are activated by winding A and B. When activating winding A, 4 of the poles will be magnetized, 2 of them with north polarity and 2 with south polarity. As shown in Figure 2, rotor teeth are aligned with the A windings and not with the B, as the sequence of steps enables B next, the rotor will align with B, therefore generating the movement as the alignment happens, doing this repeatedly is what generates stepper motor rotation.

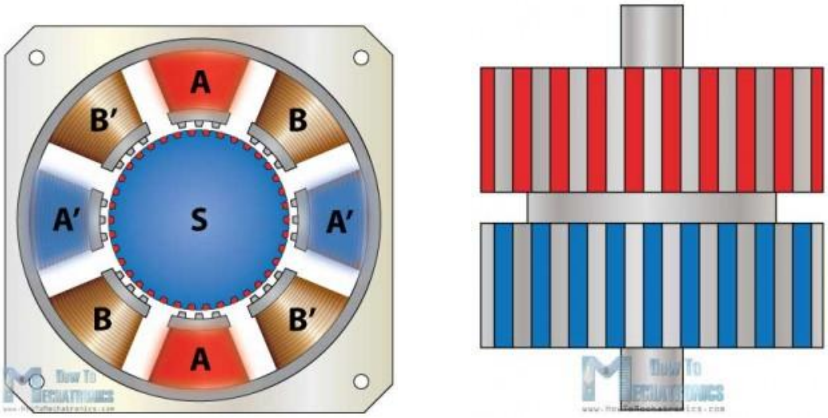


Figure 2: Toothed hybrid stepper and side view of rotor

The way these hybrid steppers work is by having half of the rotor be a north pole and half of it being a south pole as shown in Figure 2. The rotor in the provided stepper has 50 teeth, which is the most common for hybrid stepper motors. The number of teeth in the stator is different from the rotor by a small number (typically 2) to prevent both sets of teeth from fully aligning.

At each step, the variation in the alignment of teeth due to the offset between the rotor and stator, as shown in Figure 3, produces a small magnetic force that moves the motor incrementally. Referring to Figure 2, we can see that when coil A is energized, The teeth are fully aligned with the teeth in the stator A, since it acts as a north pole, and therefore attracting the teeth in the south pole of the rotor but fully unaligned with A' since the south side of the magnet in the stator is being repelled by the south pole generated by A'.

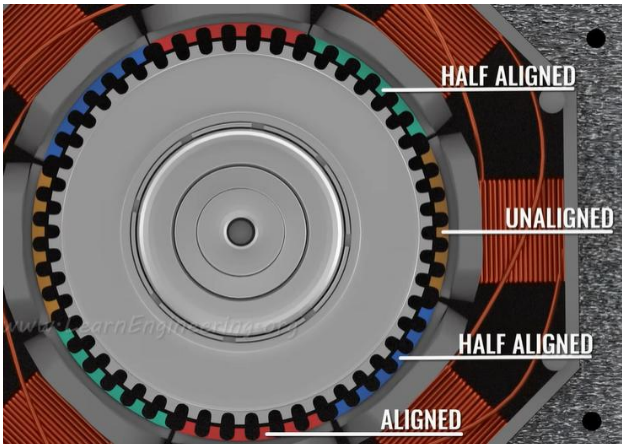


Figure 3: Rotor and stator teeth alignment

When coil B is energized, the rotor rotates a small angle to align with the teeth that were half aligned in the previous step, since that is the new north/south pole. As coil A and B are energized, they attract and repel the rotor's magnetic field constantly to generate the rotation. In the case of a stepper with 50 teeth, it takes 4 steps (A,B,A' and B') for the teeth to align with the teeth next to the starting point, therefore, every step taken moves the rotor by 1.8 degrees.

Winding Configurations

The winding configuration of a stepper motor determines how the direction of current, which determines the polarity of the electromagnetic coils, is switched. The two basic winding configurations for stepper motors are unipolar and bipolar windings (Figure 4).

Unipolar motors have 6 wires: 4 for the motor windings and a connection point, called a center tap, on each phase. The center tap allows current to flow through half of the winding at a time. Changing the polarity (commutation) only requires switching to the other section of the winding, which can be achieved simply with a single transistor for each half winding.

Bipolar motors only require 4 wires, 2 for each winding. In the case of a bipolar stepper motor, the stator has two windings that each allow bidirectional current. An H-bridge circuit is typically used to control the current flow through each winding. Bipolar stepper motors offer higher torque and better efficiency by utilizing the entire winding during operation. The circuit differences can be seen in Figure 4.

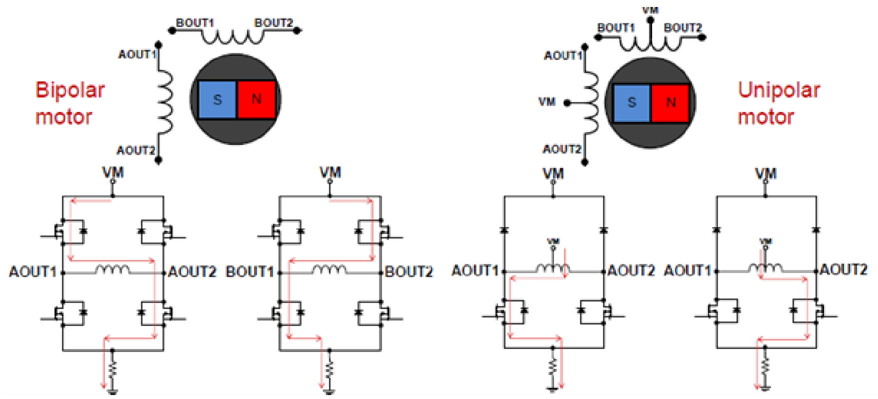


Figure 4: Unipolar and bipolar stepper motor circuits

Stepping Modes

There are several standard ways to drive the stepper motor, called stepping modes, which each affect the resolution, torque, and smoothness of the motor. We will use a simplified diagram of a permanent magnet stepper motor to simplify the descriptions. Note, however, that the working principle remains for the hybrid stepper motor and that the only difference lies in the number of steps per full rotation.

Wave Drive

In wave drive (or single-coil excitation), only one coil is energized at a time. This method minimizes power consumption but provides lower torque since only one winding is active per step. As shown in Figure 5, when coil A is energized, the south pole of the rotor magnet aligns with it. When coil B is energized next, the magnet will rotate clockwise, aligning the south pole with B. In this example, it will take 4 steps to do a full cycle rotation. For stepper motors, the step size determines how many steps it will take to do a full 360˚ rotation.

It is important to note that when we refer to energizing A′ or B′, it means applying current in the opposite direction compared to A or B. This reverses the magnetic polarity of the winding, effectively making A′ or B′ behave like a north pole when A or B would act as a south pole.

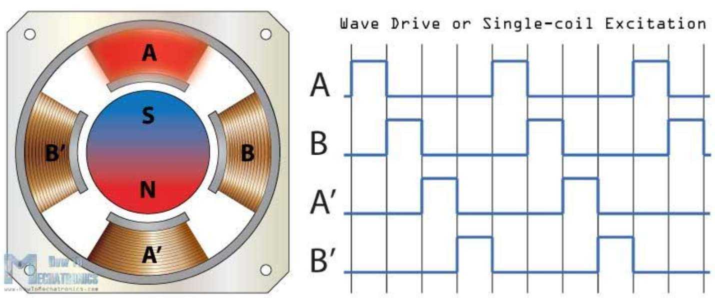


Figure 5: Stepper wave drive input sequence

Full Step Drive

To achieve the rated torque of the motor, a full step drive is used. In this excitation mode, two neighboring phases are active at any given time, resulting in the maximum rated torque and the rotor aligning with a position between the active phases (Figure 6). Compared to using wave drive, full step drive achieves a much higher torque while taking the same number of steps to complete one full rotation.

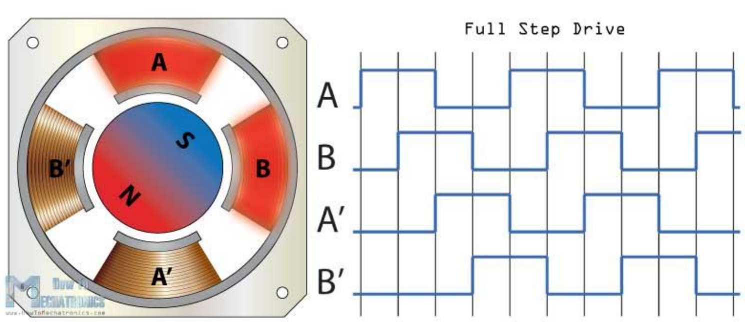


Figure 6: Stepper full step drive input sequence

Half Step Drive

Another excitation mode which combines the previous two modes is the half-step drive. Here, the motor alternates between one and two active coils, resulting in double the angular resolution of wave and full step drives. This also means that it needs double the number of steps to complete one full rotation. In this mode, the holding torque varies with each half step depending on whether one or two phases are active.

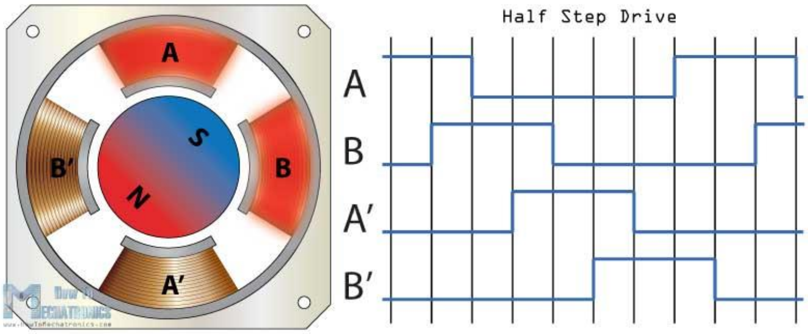


Figure 7: Stepper half step drive input sequence

Microstepping

Lastly, a stepper motor can be driven using micro-stepping, where current is applied in a sinusoidal pattern. This mode divides each step into smaller steps, allowing for increased granularity and smoothness of motor control. The most common way this mode is implemented is sine-cosine micro-stepping, depicted in Figure 8. In contrast to the other modes described above, micro-stepping requires a more complex amplifier circuitry but allows extremely fine position control. The highest torque is achieved in a holding position and decreases with an increase in speed.

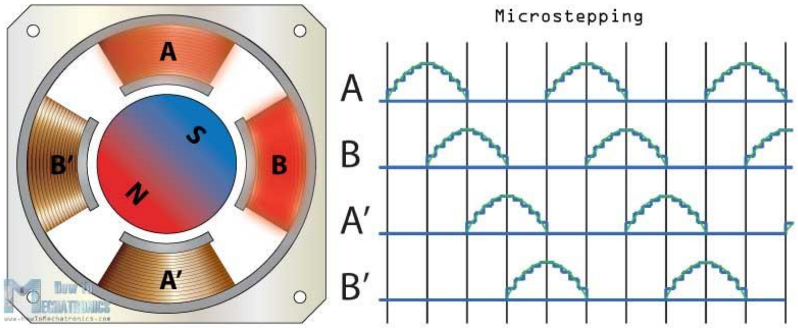


Figure 8: Microstepping input sequence

Datasheet And Considerations

The motor's datasheet summarizes its key electrical and mechanical characteristics and are essential when selecting a motor for a project or application.

Understanding a Stepper Motor Datasheet

When selecting a stepper motor for a project, key information can be found within the datasheet. Figure 9 shows two excerpts from the datasheet for the stepper motor provided with the actuators trainer.

- **Specifications**
  - Bi-polar

Series & Length	Model Number	Holding Torque		Rated Current	Resistance per Phase	Inductance per Phase	Detent Torque		Rotor Inertia	
		N.cm	oz-in	A	ohm	mH	N.cm	oz-in	g.cm²	oz-in²
PHB42S48 48mm	PHB42S48-401	52	72.8	1.3	3.2	5.5	2.6	3.64	68	0.374
	PHB42S48-402	52	72.8	1.7	1.8	3.2	2.6	3.64	68	0.374
	PHB42S48-403	52	72.8	2.3	1.2	1.6	2.6	3.64	68	0.374

Step Angle	1.8°
Step Angle Accuracy	±5% (full step, no load)

Figure 9: Stepper motor specifications

Below are some descriptions of common stepper motor specifications to consider for your application. This section will mostly focus on the electrical information, however, note that there will also be mechanical information about the motor, usually in the form of a mechanical drawing. This information is important for deciding if a motor will physically fit in a required space, or mount effectively. There may also be information on the total weight of the actuator, which may be important for applications such as serial manipulators, or model airplane control surface actuators.

Step Angle

The resolution of the motor can be provided as either the step angle or the number of steps per revolution. The requirement for resolution will vary depending on the precision and accuracy needs of the application.

Rated Voltage

This is the voltage at which the motor is designed to run, and at which the motor will have been tested for other specifications. If the motor voltage is significantly greater than that of the power supply driving it, the system will be unable to produce the rated torque. If the supply voltage exceeds the rated voltage, there is a risk of damaging the motor hardware.

Rated Current

This is the maximum current each phase can handle continuously without overheating. The peak current is drawn when the coil activation changes. However, the current drawn for the stepper will be relatively consistent as long as a coil is energized. Since stepper motors draw current even at standstill, exceeding the rated current can cause excessive heat buildup. Undersupplying current reduces available torque and may lead to skipped steps under load.

Max/Holding Torque

Holding torque is the maximum torque the motor can resist when energized but stationary. This is typically the highest torque rating for a stepper and is important when the motor must maintain position under load. If the applied load exceeds holding torque, the motor will skip steps, resulting in loss of position, which can cause errors in overall system behavior.

Pull-in/Pull-out Torque

These are specific torque values that define the maximum load of a stepper motor at a given speed without losing steps. Pull-in torque is the maximum torque under which the stepper will start or stop without acceleration. Pull-out torque is the maximum torque that can be applied to a stepper motor that is already in motion. Both values are inversely related to the speed.

Detent Torque

This is the amount of torque that an unpowered stepper motor will apply due to internal magnetic forces in reaction to an external rotation. This is the torque which causes the "cogging" sensation that occurs when rotating a stepper motor manually.

Motor Speed

For stepper motors, the maximum speed will be a function of the pull-out torque. The motor produces decreasing torque as the speed increases, shown in the speed-torque curve provided by the manufacturer. This curve is essential for selecting a motor that meets both speed and load requirements. This speed-torque curve may be given in a plot such as that shown in Figure 10, which shows the maximum speed that the motor can achieve under a given torque load.

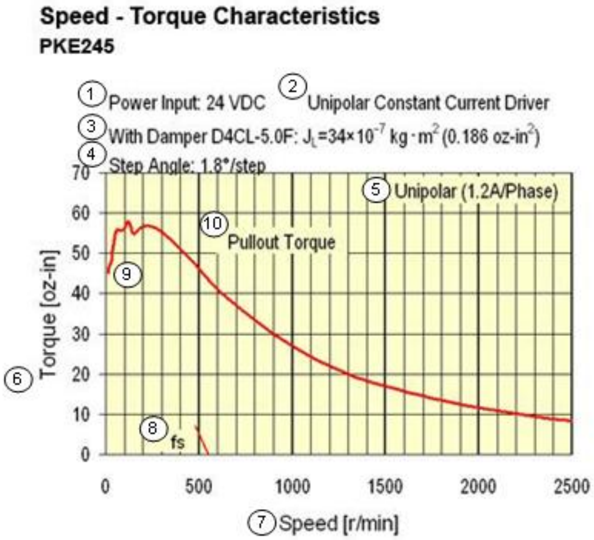


Figure 10: Speed-torque curve for a stepper motor

Design Decisions

This section describes some considerations when determining if a stepper motor is suitable for a specific application.

Cost

Stepper motors are generally cost-effective, especially for moderate- to high-torque applications. Having the capability to achieve sufficient performance with open-loop position control also eliminates the cost of position sensors. However, the required driver electronics on top of the cost of the motor itself will add to the overall system cost.

Efficiency

Stepper motors are not particularly efficient in terms of power-to-work conversion. They consume constant current even when holding position (high current) or under no load, which leads to continuous power dissipation and heat generation. Many stepper motors are designed to operate safely at elevated temperatures, often exceeding 100 ˚C, but thermal management may still be necessary in tightly enclosed systems.

Ease of Use

Controlling a stepper motor requires generating specific and precisely timed excitation signals, often through a microcontroller or dedicated driver IC. This makes setup more complex than something like a brushed DC motor. However, modern stepper drivers handle much of this complexity, providing simple interfaces for direction and step input that make integration easier.

Lifespan

Stepper motors tend to have long operational lives due to their simple, brushless construction and lack of internal feedback components.

Range of Motion

Stepper motors offer continuous 360˚ rotation and are well-suited for applications requiring unlimited or repeated motion in either direction.

Position Control Accuracy

Without built-in feedback, stepper motors can still achieve precise position control using open-loop stepping, but missed steps due to excessive loading can decrease accuracy. With techniques like microstepping, the resolution can be increased significantly but comes with the loss of holding torque.

Comparison To Servo Motors

A major advantage of stepper motors is their ability to achieve open-loop position control without requiring position sensors. This also means that stepper motors cannot detect or correct for missed steps due to overloading, unlike servo motors, which have built-in position control that allow them to return to the correct position when the load is removed. Compared to servo motors, stepper motors provide high torque at low speeds with a more significant decline as speed increases. This makes stepper motors more suitable for low-speed applications, while servo motors are better for applications that require sustained torque at high speeds.